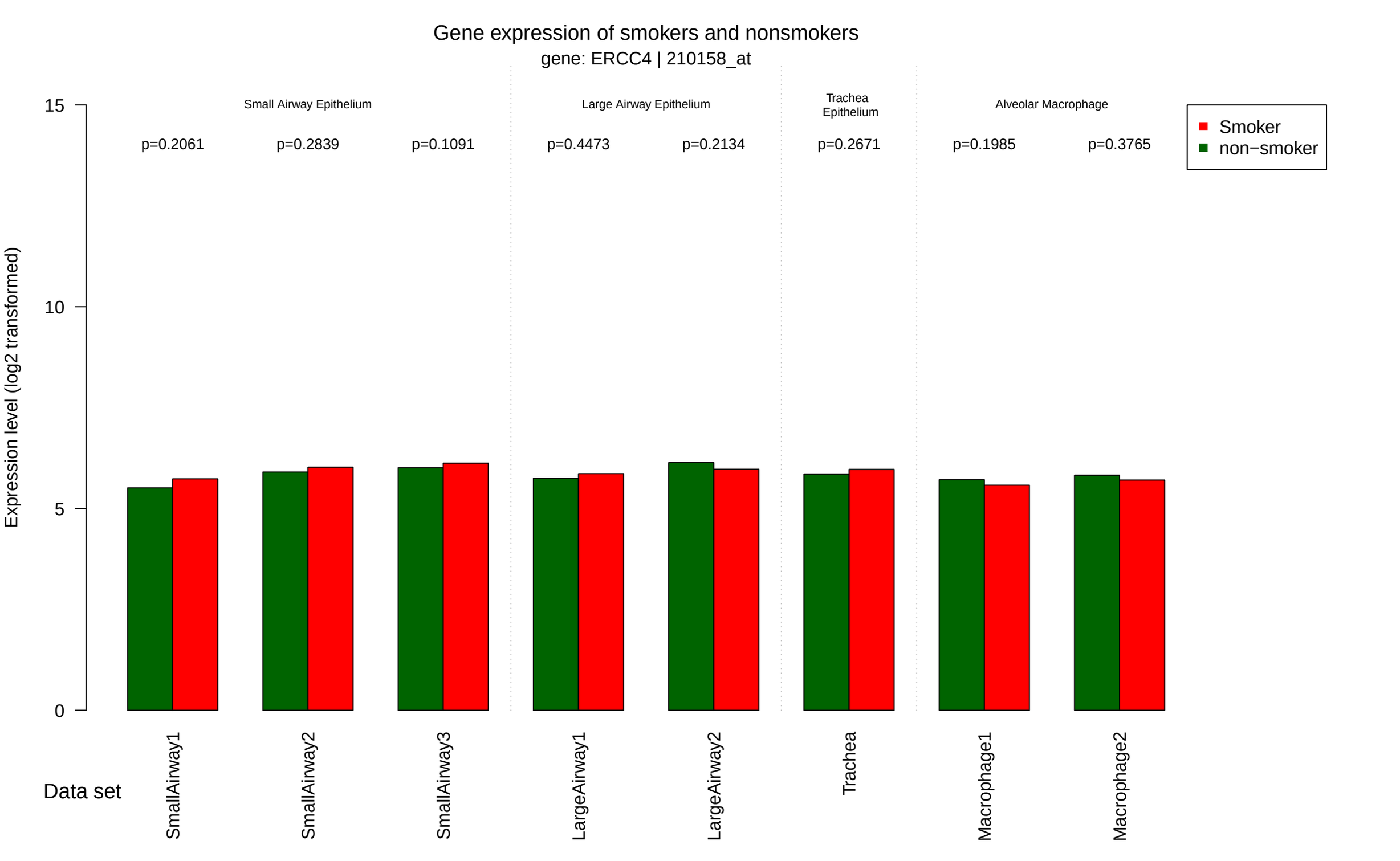
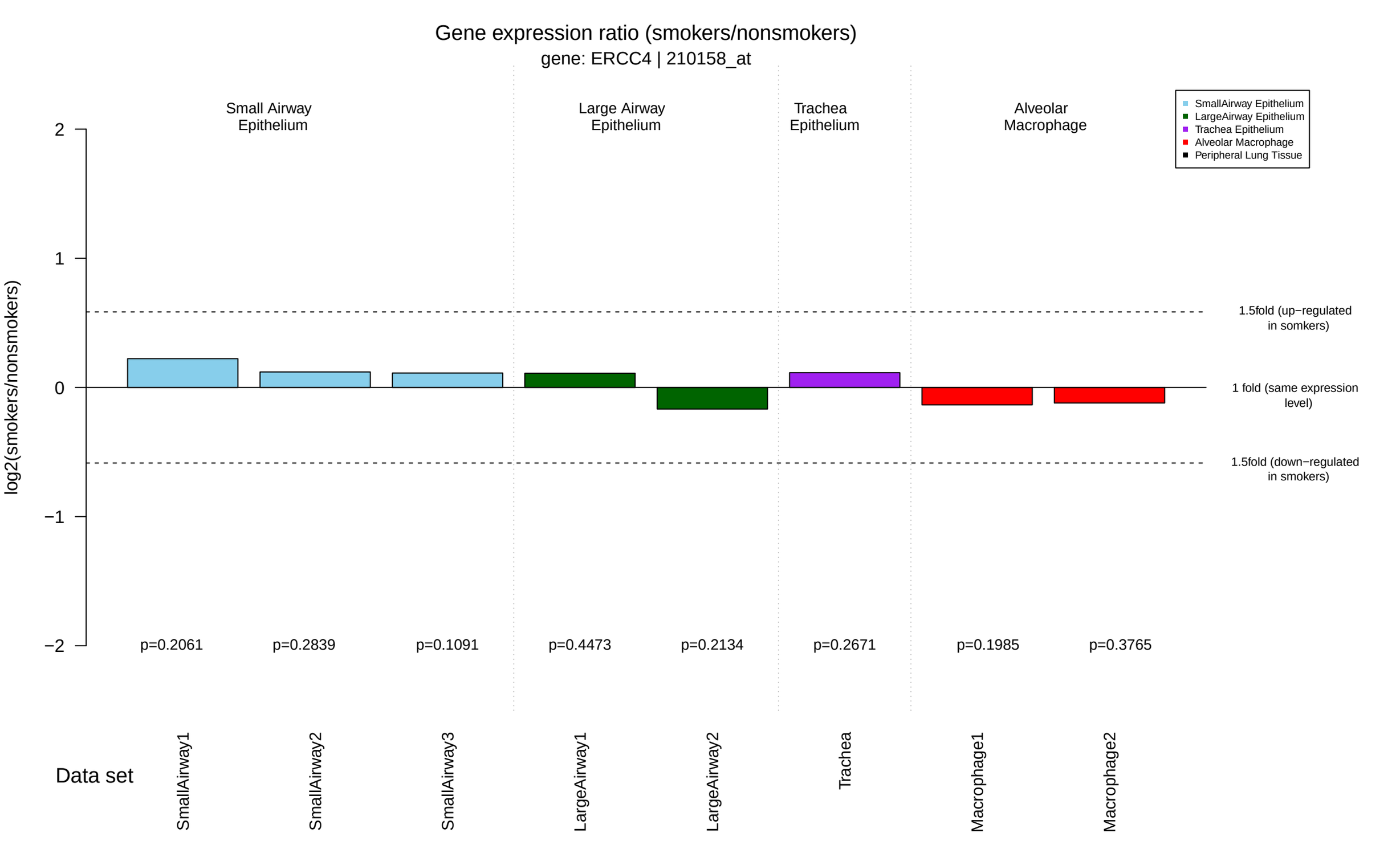
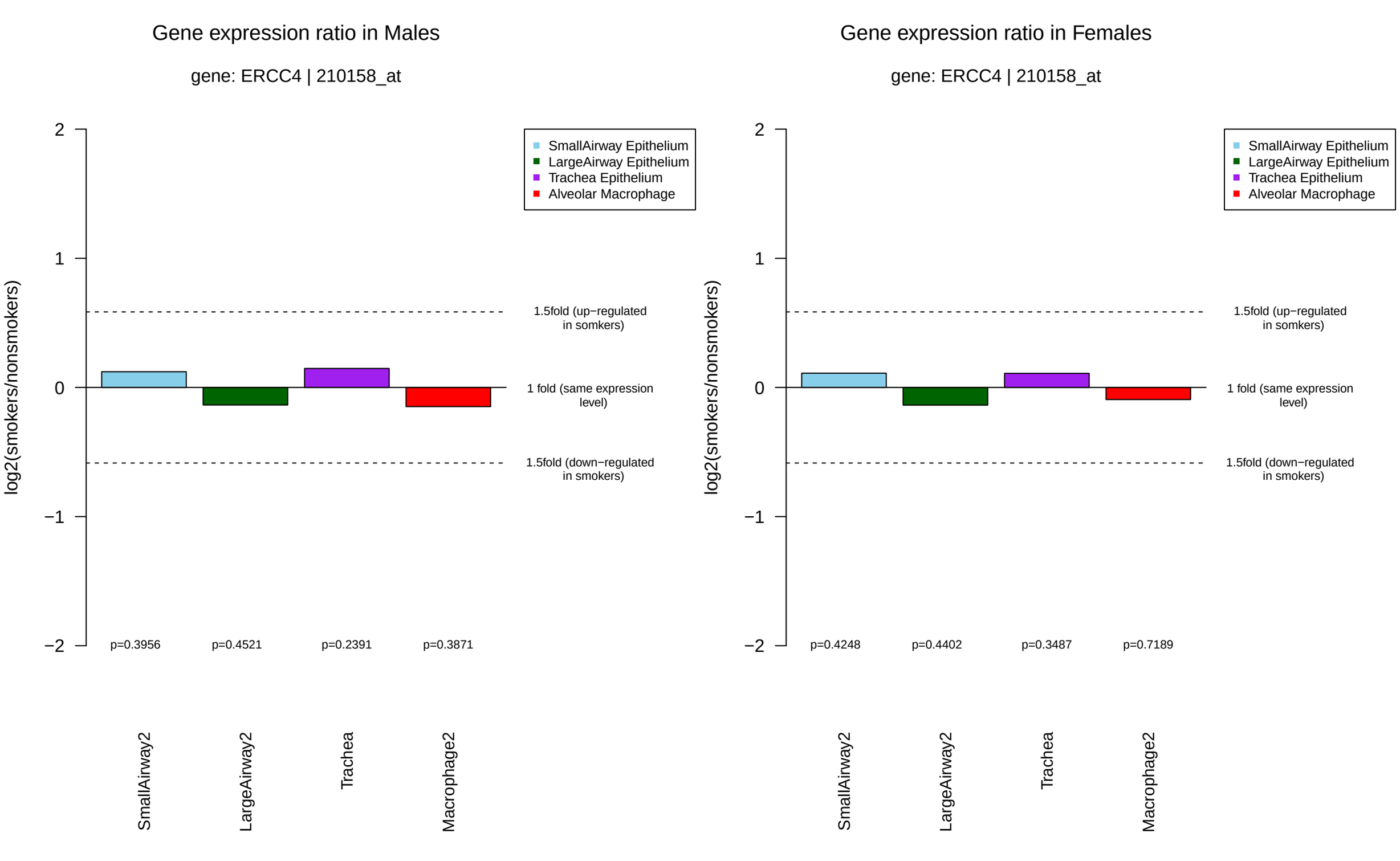
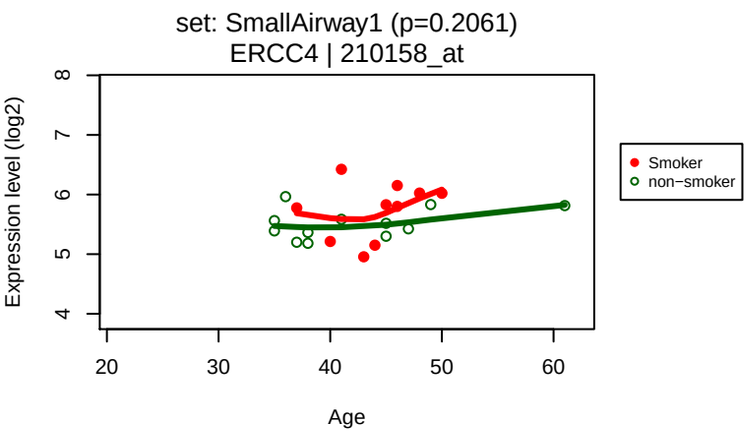
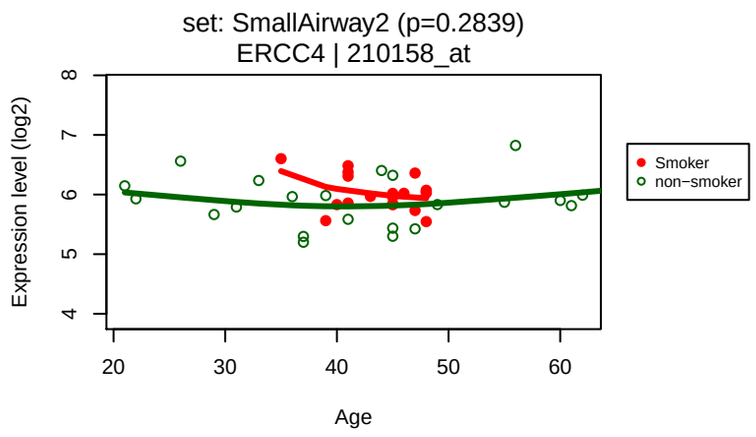
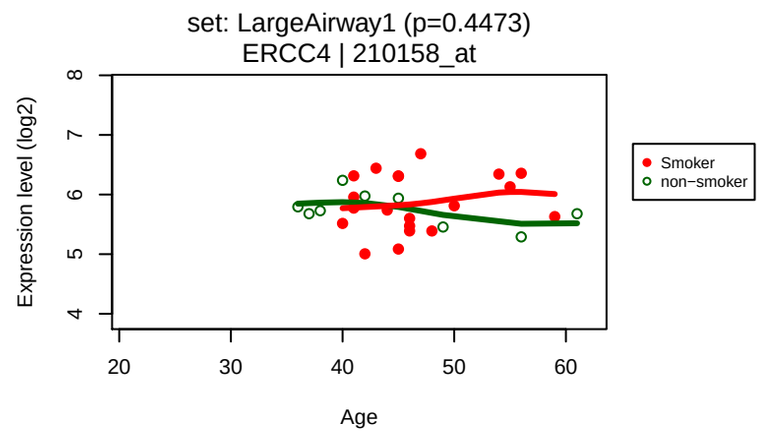

Gene expression of smokers and nonsmokers
gene: ERCC4 | 210158_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2061
p=0.2839
p=0.1091
p=0.4473
p=0.2134
p=0.2671
p=0.1985
p=0.3765
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

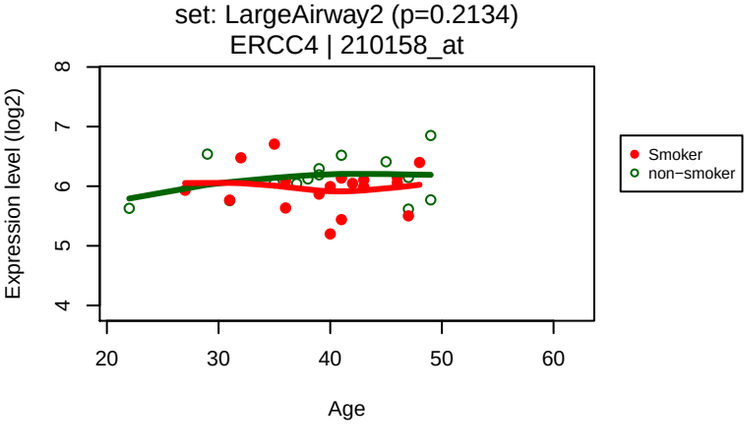

Gene expression ratio (smokers/nonsmokers)
gene: ERCC4 | 210158_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2061
p=0.2839
p=0.1091
p=0.4473
p=0.2134
p=0.2671
p=0.1985
p=0.3765
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

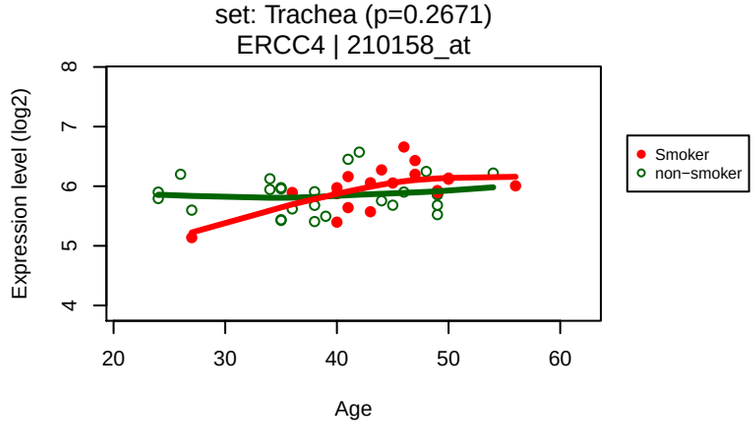

Gene expression ratio in Males
gene: ERCC4 | 210158_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3956
p=0.4521
p=0.2391
p=0.3871
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ERCC4 | 210158_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4248
p=0.4402
p=0.3487
p=0.7189
SmallAirway2
LargeAirway2
Trachea
Macrophage2

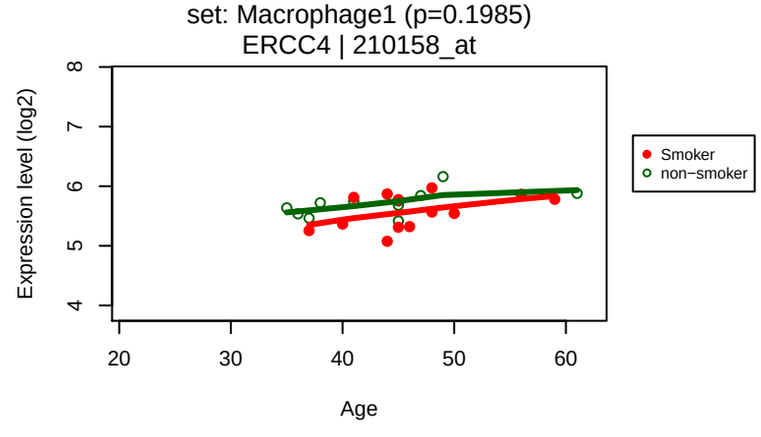

set: SmallAirway1 (p=0.2061)
ERCC4 | 210158_at
Expression level (log2)
Age
Smoker
non−smoker

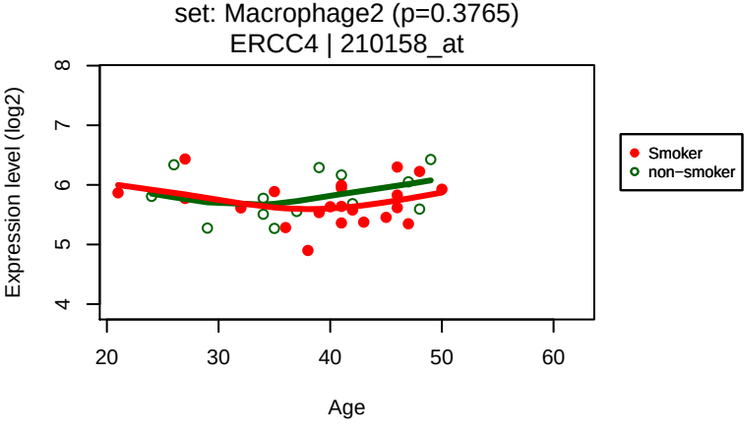

set: SmallAirway2 (p=0.2839)
ERCC4 | 210158_at
Expression level (log2)
Age
Smoker
non−smoker

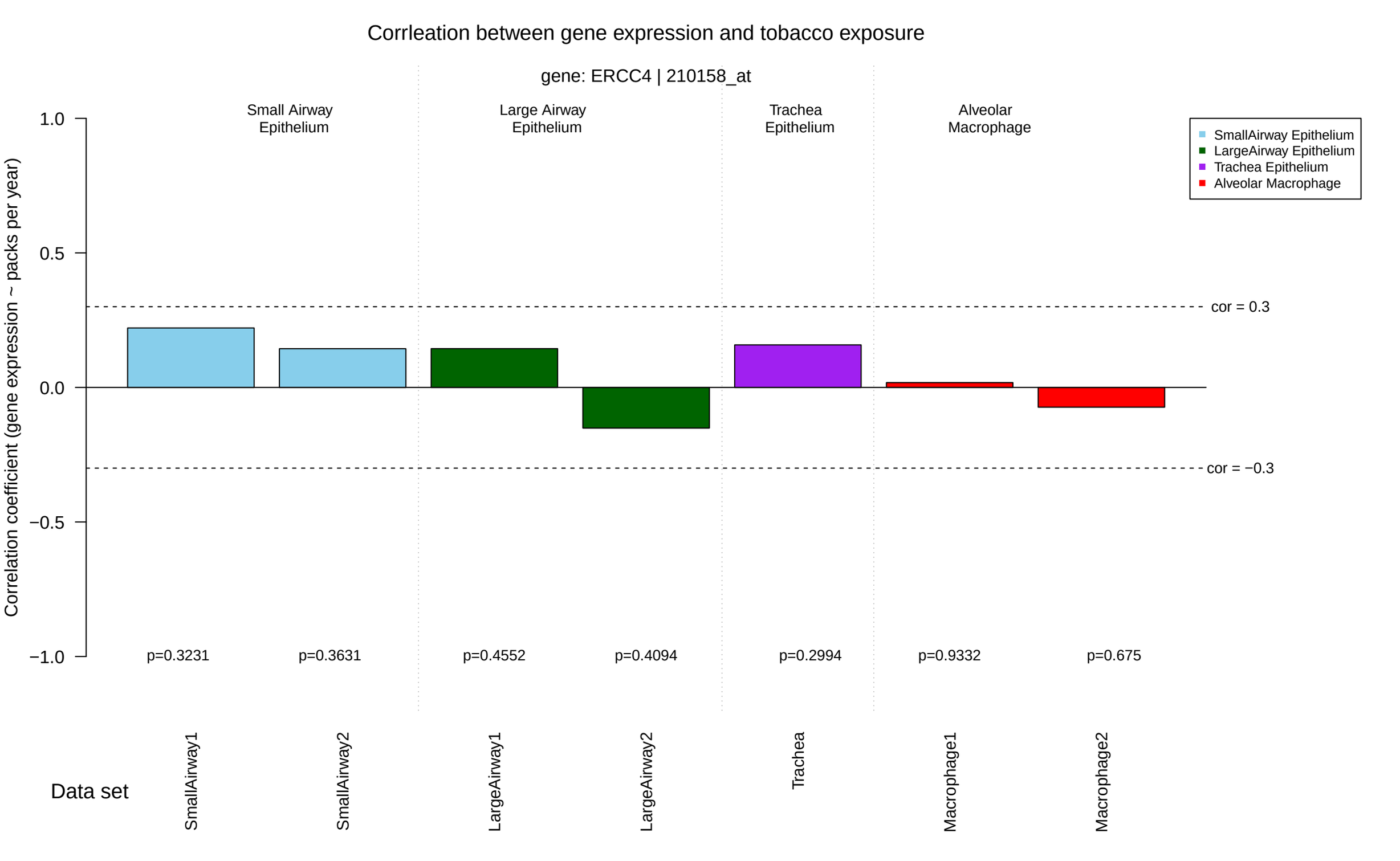

set: LargeAirway1 (p=0.4473)
ERCC4 | 210158_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2134)
ERCC4 | 210158_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2671)
ERCC4 | 210158_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1985)
ERCC4 | 210158_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3765)
ERCC4 | 210158_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ERCC4 | 210158_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3231
p=0.3631
p=0.4552
p=0.4094
p=0.2994
p=0.9332
p=0.675
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2